

Adam Zhu

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EDUCATION

Carnegie Mellon University

Bachelor of Science in Statistics and Machine Learning

Pittsburgh, PA

Expected May 2029

EXPERIENCE

SURF Research Intern, Research Data and Computing Office

Jun. 2026 – Aug. 2026

National Institute of Standards and Technology (NIST)

Gaithersburg, MD

- Built an end-to-end machine learning pipeline in Python to classify bot vs. human traffic across 190,000+ web requests, enabling accurate usage metrics for the NIST public data portal by filtering out inflated automated traffic.
- Parsed 56 MB of Apache access logs into structured data with regex and pandas at zero parse errors; engineered per-IP behavioral features (request timing, error rate, endpoint variety), excluding user-agent strings to prevent label leakage.
- Trained and compared 4 classifiers using scikit-learn pipelines with stratified splits, reaching $F1 = 0.90$ with a random forest; validated ground-truth labels by cross-checking the 25% bot rate against the 23% HTTP 429 rate-limited share.

Research Intern

Jun. 2023 – Aug. 2024

Center for Statistics and Applications in Forensic Evidence, Iowa State University

Ames, IA

- Conducted forensic data analysis on 24,000+ camera images to evaluate camera fingerprinting for device identification; co-authored a manuscript under review at *Forensic Sciences*.
- Developed methods to degrade and compare handwriting samples, creating a realistic dataset for handwriting similarity analysis.

Research Collaborator

Jan. 2024 – Jun. 2024

Virtual Reality Applications Center, Iowa State University

Ames, IA

- Conducted statistical analysis of experimental data evaluating automated communication strategies in simulated emergency evacuations; co-author of a manuscript in preparation.

PROJECTS

TrueLine — iOS Bowling Ball Tracker | *Swift, SwiftUI, Core ML, Python, PyTorch, OpenCV*

2025 – Present

- Developing an iOS app (App Store launch in preparation) that turns a single iPhone into a ball-motion tracker, computing per-throw launch speed, breakpoint, and entry angle — metrics otherwise requiring a \$10k+ in-lane installation.
- Built an on-device computer vision pipeline with zero third-party dependencies: fine-tuned a YOLOv8 ball detector on hand-labeled footage, exported to Core ML, with Kalman-filter tracking and Savitzky–Golay path smoothing.
- Mapped arbitrary camera angles into real-world lane coordinates via homography from a 4-corner calibration, auto-proposed by a custom gutter-line detector and refined through a magnifier-loupe drag UI.
- Verified the Swift port against a Python/OpenCV prototype using a clip-by-clip parity harness — primary metrics agree within ~1 board and launch speed within 1–2%.

PUBLICATION

Lin, H., **Zhu, A.** and Wang, C. (2024). Statistical Tests for Proportion Difference in One-to-Two Matched Binary Diagnostic Data: Application to Environmental Testing of Salmonella in the United States. *Mathematics* 2024, 12, 741. doi.org/10.3390/math12050741

LEADERSHIP

Co-Captain

Jun. 2021 – Jun. 2025

Team Neutrino, FIRST Robotics Team #3928

Story County, IA

- Led 7 sub-teams and 40+ members in outreach, fundraising, and operations; represented the team to sponsors and judges, contributing to multiple NASA Engineering Inspiration Awards and World Championship qualifications.

Camp Co-Designer

Summers 2024 & 2025

COR Robotics Summer Camps

Central Iowa

- Co-developed robotics and engineering curriculum (drone racing, battle drones) and taught 100+ students in grades 3–8 foundational STEM concepts through project-based learning.

TECHNICAL SKILLS

Languages: Python, Swift, R, Java, MATLAB, $\LaTeX$

Frameworks & Libraries: pandas, scikit-learn, PyTorch, OpenCV, FastAPI, SwiftUI, Core ML, AVFoundation

Tools & Data: Git/GitHub, MongoDB, REST APIs, JSON/JSON-LD, regex

Statistics & ML: Classification, Feature Engineering, Model Evaluation, Experimental Design, GLMs, ANOVA

HONORS & AWARDS

Congressional Award (Silver Certificate), The Congressional Award Foundation, U.S. Congress

Jan. 2025

1st Place — Mathematics Category & RAYGUN Innovative Scientist Award, SSTFI

Apr. 2024

Eagle Scout Rank, Boy Scouts of America

May 2023

World Championship Engineering Inspiration Award, FIRST Robotics (1 of 6 in the world)

Apr. 2022